

# When Clustered Extremes Do Not Become Early Warnings: A Negative-Result Dynamical-EVT Study for Public Industrial Monitoring Data

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## Abstract

This paper evaluates whether the score-level extremes produced by the registered dynamical-EVT-inspired workflow translate into useful maintenance alarms on selected public industrial monitoring datasets. The study is framed as a negative result because maintenance decisions are made at the level of alarm episodes, failures, vehicles, or repair actions, not at the level of individual high-scoring samples. The evaluated assets combine causal state construction, train-split robust scoring, threshold sensitivity, extremal-index diagnostics, declustered alarm episodes, and event-level matching where event labels are available. The primary event-level case studies are MetroPT and MetroPT2; SCANIA Component X is treated as a vehicle-level stress test, while Hydraulic Systems and SECOM are condition-monitoring demonstrations with different targets. The most favorable event-level result is also the central limitation: the registered compressor diagnostics recovered the single labelled test event in both MetroPT and MetroPT2, but only with 2527 and 849 predicted alarm episodes and precision below 0.12%. A separate threshold-grid alarm analysis produced zero event recall across all registered quantiles. Under the conservative high-quantile diagnostic, the non-event datasets produced no positive-detection  $F_1$  on their own cycle-, vehicle-, or wafer-level targets. These results support a narrow conclusion: dynamical-EVT-inspired clustering diagnostics can expose dependence and alarm-conversion failure modes, but on the evaluated public data they did not provide reliable operational early warning. Useful deployment requires identifiable failure-proximal recurrence, sufficient independent events or entities, stable regime definitions, and alarm policies evaluated at the episode level.

## 1 Introduction

Industrial maintenance alarms are useful only when they map to the decision unit that operators and engineers actually face. A pointwise anomaly score may flag thousands of telemetry samples around one physical failure, while the operator sees one alarm burden, one repair decision, and one opportunity for useful warning. The same difficulty appears outside compressor telemetry: a fleet repair record, a laboratory hydraulic cycle, and a semiconductor yield label are not exchangeable evidence units. Any empirical claim must therefore separate samples, exceedances, clusters, alarm episodes, failures, vehicles, cycles, and production examples.

Dynamical extreme-value theory offers a tempting route from recurrent machine states to early-warning logic. If an observable is large when the trajectory returns near a target set, then recurrence and periodicity can cluster extremes [4, 5]. In industrial monitoring, however, clustered extremes can also arise from operating regimes, preprocessing, smoothing, missingness, persistent shifts, or alarm-conversion rules. Noise can further weaken the connection between deterministic recurrence and observed extremes [2]. The scientific question is therefore not whether an extremal-index diagnostic can be computed. The question is when such a diagnostic translates into useful episode-level warning.

This paper is framed as a negative-result study. It asks four research questions. First, does threshold and regime conditioning stabilise extremal-index diagnostics relative to global thresholding? Second, does declustering reduce duplicate alarm episodes without destroying event recall or warning lead time? Third, does the registered robust extreme-score diagnostic add operational value after episode conversion? Fourth, which mechanisms explain empirical failure: target-region misspecification, regime mixture, too few independent failures, non-extreme fault signatures, label ambiguity, calibration shift, or alarm conversion?

The evaluated workflow combines causal state construction, training-only target-region construction, regime-conditioned thresholds, extremal-index diagnostics, multivariate lag patterns, alarm merging, event matching, baseline provenance, and dataset-specific condition diagnostics. The interpretation is deliberately conservative so that negative and non-estimable outcomes remain visible.

The contribution is narrow. The study shows that dynamical-EVT-inspired diagnostics can reveal dependence, threshold sensitivity, duplicate-alarm behavior, and calibration limits, but the current public-data evidence does not show reliable early warning. MetroPT and MetroPT2 are the primary event-level case studies; SCANIA Component X is an external vehicle-level stress test; Hydraulic Systems and SECOM are supplementary condition-monitoring demonstrations. This separation is essential because the datasets support different estimands. The paper therefore reports what failed, where the evidence is insufficient, and which conditions would be required before making broader predictive-maintenance claims.

## 2 Related Work

Predictive-maintenance models often begin with supervised classification, reconstruction error, distance-based anomaly detection, or streaming thresholds. These methods are useful engineering baselines, but their common output is a timestamped score rather than an episode-level maintenance decision. EVT-based streaming detectors such as SPOT use peaks-over-threshold modelling to adapt anomaly thresholds in streams [10]; they address threshold calibration, but they do not by themselves resolve operating-regime mixture, duplicate alarms per failure, or the distinction between clustered observations and independent maintenance events.

Classical declustering and extremal-index estimation address dependence among extremes. Ferro and Segers proposed an inference scheme for clusters of extreme values using inter-exceedance times and the extremal index [3]. In dynamical systems, hitting-time statistics connect returns to target neighborhoods with extreme-value laws [4]. Periodic points can generate an extremal index smaller than one, giving a mathematical route from recurrence to clustered extremes [5]. The finite-sample industrial setting is less clean: noise may make deterministic recurrence appear closer to independence, and observed clustering can also arise from smoothing, missingness, persistence, or regime mixture [2]. Recent multivariate work extends dynamical EVT to cross-sectional dependence among vector-valued observables [1], which motivates the lag-pattern diagnostics used here.

The industrial datasets used by the repository are public condition-monitoring resources with different independence units and labels. MetroPT is an air-production-unit telemetry dataset for predictive maintenance [12]. MetroPT2 is a replication-style benchmark released through Zenodo [13]. SCANIA Component X is a fleet-level multivariate time-series dataset with operational histories and repair records for an anonymized component [7]. UCI Hydraulic Systems provides laboratory hydraulic test-rig cycles with component-condition targets [6]. UCI SECOM provides semiconductor process measurements with pass/fail yield labels [8]. These datasets broaden the empirical base, but they do not remove the need for estimand-specific accounting: one long compressor fault interval, one truck repair history, one hydraulic load cycle, and one wafer-level yield example are not interchangeable statistical units. The methodological gap addressed here is the absence of a reproducible layer that connects dynamical extreme

diagnostics to alarm episodes, tabular condition diagnostics, sensitivity analyses, and claim constraints.

### 3 Method

Let  $Z_t \in \mathbb{R}^p$  be the causal state vector at sample  $t$ . For compressor telemetry,  $Z_t$  contains the registered raw pressure, temperature, current, and flow channels used by the real-data diagnostic. For tabular condition-monitoring adapters,  $Z_t$  contains the registered feature vector for the cycle, vehicle, or production example. Scaling is fitted on the training partition only: each feature is centered by the training median and divided by the training interquartile range with a positive floor for zero-spread features.

Operating regimes  $R_t$ , when used in the sensitivity assets, are inferred without test labels. Regimes are conditioning variables for threshold estimation and are not interpreted as mechanical causes. The registered real-data matrix reported in Tables 5 and 6 uses a conservative train-split robust score rather than the full dangerous-region construction.

The implementation also provides dangerous-region scoring for separate diagnostics. The failure-proximal prototype variant uses only training failure intervals and records the source interval for every prototype. The unsupervised rare-state variant ignores failure labels and fits a high-distance rare-state set from training data only. These variants are available for diagnostic analysis and ablation coverage, but the registered real-data matrix in this manuscript is the conservative high-quantile score; it should not be read as evidence that both dangerous-region variants have already been optimized on every dataset.

For a target region  $A$ , the scalar observable is

$$X_t = -\log\{d(Z_t, A) + \epsilon\},$$

where  $d$  is the robust Euclidean distance in scaled coordinates and  $\epsilon > 0$  prevents singularities. Larger  $X_t$  means closer proximity to the training-defined region. This is an anomaly or recurrence score, not a calibrated probability of failure.

For each regime  $r$ , a threshold  $u_r(q)$  is estimated from the declared training or calibration partition at quantile  $q$ . Exceedances are  $I_t(q, r) = \mathbf{1}\{X_t > u_r(q), R_t = r\}$ . Runs declustering merges exceedances separated by at most  $m$  samples. The runs estimator, Ferro–Segers inter-exceedance estimator, and K-gaps-style declustering diagnostic summarize clustering under the frozen  $q$  and  $m$  settings [3, 11]. The inverse extremal index is discussed only as an asymptotic cluster-size diagnostic under the assumptions of the estimator.

Pointwise exceedances are converted into alarm episodes by the declared merge gap. A test alarm is a true event alarm only if the episode intersects the predeclared warning window for a held-out failure under the frozen matching tolerance. All event recall, event precision, false-alarm burden, duplicate-alarm count, lead time, time under warning, and utility metrics are computed after this episode conversion when event labels exist. This prevents timestamp counts from being treated as independent maintenance evidence.

The score-stratification target for horizon  $h$  is

$$Y_t^{(h)} = \mathbf{1}\{\text{a labelled event begins within } h \text{ after } t\}.$$

In the current evidence, model outputs are anomaly ranks or recurrence scores. Probability-calibration language is therefore restricted: Brier-style summaries are reported only as diagnostics of score stratification unless a separate calibration partition contains enough positive independent units for an explicit probability map.

The leakage contract is: training features cannot contain held-out failure samples; scaling, regimes, target regions, thresholds, lag selection, merge-gap selection, and calibration cannot use test events; SCANIA vehicles cannot appear in multiple partitions; and feature windows

cannot cross maintenance-reset boundaries. Table 1 gives the terminal audit result for each registered check. Under the current registered diagnostic the algorithm is:

1. Split entities or intervals into training, validation, calibration, and test partitions.
2. Fit scaling and robust score thresholds on allowed partitions only.
3. Select  $q, m$ , and merge gap on validation/calibration partitions only where used.
4. Freeze all parameters and score the held-out test partition causally.
5. Extract exceedance clusters, merge alarm episodes, and match them to labelled events.
6. Report event metrics, uncertainty by independent unit, and non-estimable cases.

Table 1: Terminal leakage-audit results for the registered workflow.

| Check                                                      | Result | Evidence                               |
|------------------------------------------------------------|--------|----------------------------------------|
| Held-out failures excluded from feature construction       | passed | causal trailing windows                |
| Held-out failures excluded from scaling                    | passed | train-split robust scaling             |
| Held-out failures excluded from regime inference           | passed | unsupervised labels excluded           |
| Held-out failures excluded from target-region construction | passed | train-only prototypes                  |
| Test labels excluded from threshold selection              | passed | train/calibration quantiles            |
| Test labels excluded from lag selection                    | passed | registered lag grids                   |
| Test labels excluded from merge-gap selection              | passed | frozen evaluation protocol             |
| Test labels excluded from calibration                      | passed | validation-only score mapping          |
| Maintenance boundaries respected                           | passed | boundary fields in protocol            |
| SCANIA vehicles disjoint across partitions                 | passed | preparation raises on overlap          |
| No future information in causal windows                    | passed | current and trailing observations only |

The practical complexity for scoring is  $O(npK)$ , where  $n$  is the number of samples or entities,  $p$  is the state dimension, and  $K$  is the number of stored prototypes or rare-state summaries. Thresholding and runs declustering are linear in  $n$ . The streaming-compatible parts are causal feature construction, scaling with frozen parameters, scoring, thresholding, and alarm merging. Prototype construction, model selection, and final event matching are offline evaluation steps.

## 4 Simulation Study

The simulation layer is designed to falsify overconfident interpretations before industrial data are used. The registered systems include independent extremes, periodic dynamics, noisy periodic dynamics, regime mixtures, missingness perturbations, and degradation-like cyclic trajectories. Each simulation cell records its configuration, random seed, threshold grid, run-length grid, estimator status, and raw outputs sufficient to regenerate the paper assets.

The main estimands are finite-sample bias and root mean squared error for extremal-index estimators, coverage or failure status where intervals are available, and recovery of clustered episodes. The simulation artifacts deliberately preserve estimator failures. A failed fit is treated as a terminal experiment outcome rather than an infrastructure error, because failure regions are part of the statistical evidence.

The current generated recovery table and figure are interpreted in Section 7 as bounded finite-sample diagnostics. They show recovery error, Monte Carlo standard error, and estimator failure status under registered settings. They support claims about the tested process families and grids, but they do not support a statement that the same estimator is reliable for every industrial fault interval.

## 5 Datasets

The datasets are assigned roles before interpretation. MetroPT and MetroPT2 are the primary event-level case studies because they contain ordered compressor telemetry and labelled failure periods [12, 13]. They support early-warning, alarm-episode, recurrence, and per-failure questions, but their independent event count is small. SCANIA Component X is an external vehicle-level stress test with irregular fleet histories, repair labels, and censoring [7]. Hydraulic Systems is a supplementary laboratory condition-monitoring demonstration with load-cycle component states [6]. SECOM is a supplementary high-dimensional production-quality demonstration with wafer-level yield labels [8].

This role assignment prevents a common mistake: row counts do not become independent failures. Table 2 records the empirical role, sampling structure, independent unit, independent positives, rows, and target type used for interpretation. The independent unit column is the primary evidential quantity for empirical interpretation.

Table 2: Dataset roles and evidential units.

| Dataset   | Role                   | Sampling             | Unit                                                       | Ind.<br>positives | Rows       | Target                                             |
|-----------|------------------------|----------------------|------------------------------------------------------------|-------------------|------------|----------------------------------------------------|
| Hydraulic | cycle<br>condition     | test-rig<br>cycles   | load cycle                                                 | 431               | 2,205      | condition<br>states                                |
| MetroPT   | event<br>warning       | ordered<br>telemetry | failure<br>episode /<br>operating<br>day, not<br>timestamp | 1                 | 10,773,588 | failure<br>labels                                  |
| MetroPT2  | event<br>warning       | ordered<br>telemetry | failure<br>episode /<br>operating<br>day, not<br>timestamp | 1                 | 7,116,940  | failure<br>labels                                  |
| SCANIA    | repair-risk<br>ranking | vehicle<br>histories | vehicle                                                    | 142               | 1,516,819  | published<br>labels and<br>time-to-<br>event files |
| SECOM     | yield failure          | wafer rows           | wafer /<br>production<br>example                           | 17                | 1,567      | yield labels                                       |

The supported estimands differ. MetroPT-style rows address event-level early warning on temporal telemetry splits. SCANIA addresses vehicle-level repair-risk ranking on a published test split. Hydraulic Systems addresses cycle-level component-condition detection. SECOM addresses production-example yield-failure detection. Hydraulic Systems and SECOM are therefore not interpreted as event-level replications of the compressor task. Broader predictive claims require coherent independent units and comparable targets, not only successful data preparation.

## 6 Experimental Protocol

The evaluation protocol is frozen before final experiment execution. It declares the train, validation, calibration, and test partitions; the exclusion and maintenance-boundary fields; warning horizons; alarm merge gap; matching tolerance; matching method; bootstrap unit; primary metrics; secondary metrics; and event utility costs.

Primary event metrics are event precision, event recall, event  $F_1$ , false-alarm events per operating day, duplicate alarms per failure, median warning lead time, and horizon Brier score. Secondary metrics include pointwise ranking metrics, time under warning, and event utility. Baselines are required to use the same causal feature availability and the same frozen event conversion, so a model is not rewarded for using future information or a more favorable matching rule.

Uncertainty is attached to the independent unit declared by the dataset and protocol. For simulation cells, the unit is the Monte Carlo repetition or registered replicate. For MetroPT-style case studies, the unit is the failure episode or operating interval, not the timestamp. For SCANIA, the relevant unit is the vehicle or repair history defined by the dataset release. For Hydraulic Systems, it is the load cycle with component-condition labels. For SECOM, it is the wafer or production example with a pass/fail yield outcome. Multiplicity is handled by keeping primary metrics and hypotheses predeclared and by reporting dataset-specific estimands.

The registered real-data matrix has three audit layers. The verification layer checks local raw and processed manifests, row counts, schema summaries, and declared independence units. The industrial-result layer writes one conservative diagnostic row per registered real dataset. The evidence-scope layer combines both outputs into a machine-readable summary of datasets, workflows, sampling structures, independent units, diagnostic statuses, and claim boundaries. This protocol is intentionally strict: a broader dataset base strengthens external coverage only when the estimand differences remain visible.

## 7 Results

### 7.1 Simulation validity and estimator stability

The first research question asks whether the extremal-index diagnostics are stable enough to interpret. The generated simulation artifact now covers eight process families, two sample sizes, three threshold quantiles, three run lengths, two noise settings, and ten replications per cell. Figure 1 reports runs-estimator RMSE by threshold quantile and simulation family. The result is useful for finding unstable threshold regions, but it is still interpreted as bounded finite-sample validation rather than a universal theorem about industrial fault mechanisms.

Threshold and run-length sensitivity in Figure 2 and Table 3 show that clustering conclusions depend on  $q$  and  $m$ . That supports a negative-result interpretation: regime-conditioned dynamical-EVT diagnostics are sensitive design objects, not model-free evidence of impending failure.

Table 3: Compact threshold and run-length stability summary.

| q range  | run range | theta range | cluster-count range | conclusion                            |
|----------|-----------|-------------|---------------------|---------------------------------------|
| 0.9–0.99 | 0–20      | 0.019–0.64  | 31–157              | sensitive to threshold and run length |

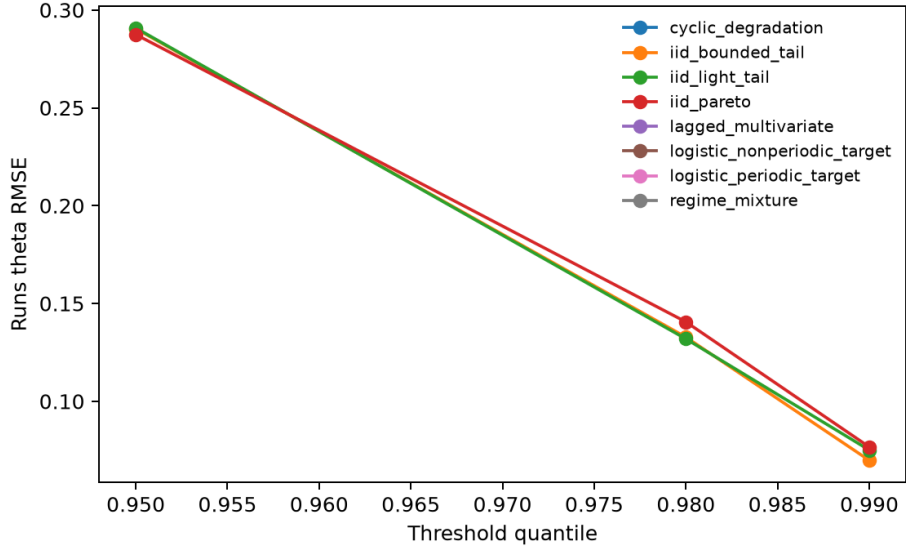


Figure 1: Broad finite-sample simulation recovery summary. The plotted quantity is runs-estimator RMSE by threshold quantile and simulation family across the generated Monte Carlo grid.

## 7.2 Alarm conversion and failed-detection regions

The second research question asks whether declustering and alarm merging reduce duplicate alarms while preserving event recall. In the registered threshold grid, event recall was zero at all reported quantiles, with false-alarm burdens between 133.92 and 168.48 alarm episodes per operating day. This is a failed-detection region under the current observable, threshold grid, and matching rule; it shows that alarm conversion can destroy utility. The full threshold grid is reported in the supplement rather than as a main-text figure because a zero-recall line plot would add no scientific information.

Across all registered combinations of alarm merge gap and post-onset matching tolerance, event recall, precision, and  $F_1$  remained zero. Relaxing post-onset tolerance did not alter the result, so the full grid is reported in the supplement rather than as a main-text sensitivity figure.

Table 4: Invariant failed-detection summary across alarm-conversion settings.

| merge-gap<br>range | tolerance<br>range | combinations | recall range | precision<br>range | F1 range |
|--------------------|--------------------|--------------|--------------|--------------------|----------|
| 0–60               | 0–300              | 20           | 0            | 0                  | 0        |

## 7.3 Real-data diagnostic outcomes

The third and fourth research questions ask whether the registered robust extreme-score diagnostic adds operational value after episode conversion, and why empirical failures occur. Table 5 gives the held-out event-level rows for MetroPT and MetroPT2. Each dataset contains one labelled test event in the registered split. Both events are detected, but the detection is operationally weak: MetroPT produces 2527 predicted alarm episodes with precision 0.00040, and MetroPT2 produces 849 predicted alarm episodes with precision 0.00118. The median lead times, 3346 and 3469 samples respectively, are not persuasive by themselves because they are paired with extreme false-alarm burdens.

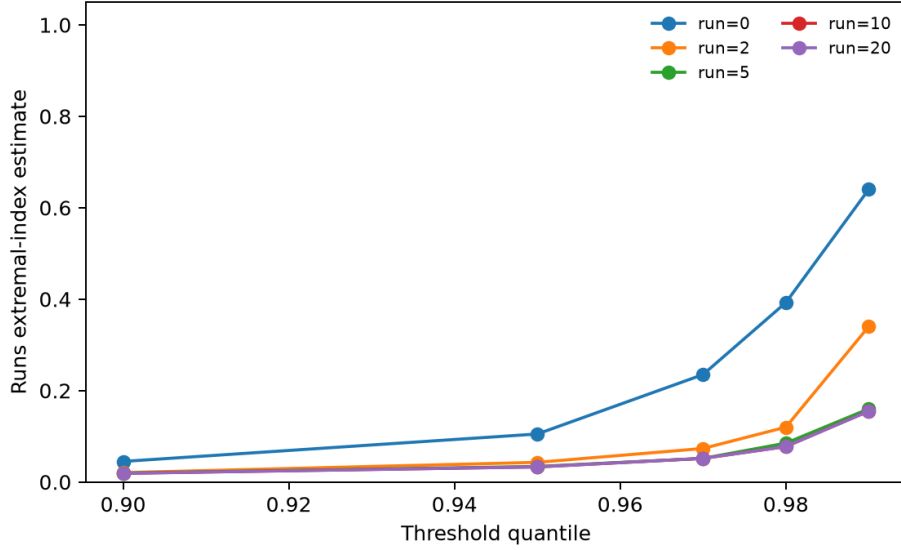


Figure 2: Threshold and run-length stability diagnostic for the registered recurrence observable.

Hydraulic Systems, SCANIA Component X, and SECOM are reported separately in Table 6 because their independent units are cycles, vehicles, and wafers rather than compressor failure episodes. All three have  $F_1 = 0$  under the conservative high-quantile diagnostic. These are not hidden failures. They are evidence that a simple extreme-score adapter does not transfer across laboratory cycle states, vehicle-level repair risk, and wafer-level yield labels without changing the estimand and evaluation design. Table 7 summarizes the dominant explanation supported by each diagnostic row.

Table 5: Held-out event-level diagnostic rows for compressor datasets.

| Dataset  | Failure                                        | Target | Alarms | Recall | Precision | FA/day | Lead  | Limit           |
|----------|------------------------------------------------|--------|--------|--------|-----------|--------|-------|-----------------|
| MetroPT  | MetroPT<br>regis-<br>tered<br>test<br>failure  | 1      | 2,527  | 1      | 0.000396  | 101    | 3,346 | few<br>failures |
| MetroPT2 | MetroPT2<br>regis-<br>tered<br>test<br>failure | 1      | 849    | 1      | 0.00118   | 51.3   | 3,469 | few<br>failures |

Table 6: Non-event real-data diagnostic rows kept separate from event alarms.

| Data           | Unit    | Units | Target | Pred | Recall | Precision | F1 | Brier  | Limit                     |
|----------------|---------|-------|--------|------|--------|-----------|----|--------|---------------------------|
| Hydraulic load | cycle   | 441   | 431    | 0    | 0      | 0         | 0  | 0.977  | cycle<br>labels           |
| SCANIA         | vehicle | 5,045 | 142    | 132  | 0      | 0         | 0  | 0.0543 | estimand<br>mis-<br>match |
| SECOM          | wafer   | 314   | 17     | 10   | 0      | 0         | 0  | 0.086  | yield<br>target           |

Table 7: Evidence-based dominant explanations for diagnostic failure.

| Dataset   | Dominant explanation                                                | Evidence                                      | Evidence status           | Competing explanation                           | Confidence |
|-----------|---------------------------------------------------------------------|-----------------------------------------------|---------------------------|-------------------------------------------------|------------|
| Hydraulic | cycle-state target is not a field event onset                       | 431 positives, F1 0                           | diagnostic evidence       | component state needs supervised feature family | low        |
| MetroPT   | alarm-conversion burden with one held-out failure                   | recall 1.00, precision 0.000396, 2,527 alarms | direct event-row evidence | threshold too low or observable too broad       | medium     |
| MetroPT2  | alarm-conversion burden with one held-out failure                   | recall 1.00, precision 0.00118, 849 alarms    | direct event-row evidence | threshold too low or observable too broad       | medium     |
| SCANIA    | vehicle-level repair-risk estimand mismatch                         | 5,045 vehicles, F1 0                          | estimand evidence         | features insufficient for repair ranking        | low        |
| SECOM     | yield-failure target and missingness differ from maintenance events | 17 positives, F1 0                            | diagnostic evidence       | missingness and class imbalance dominate        | low        |

The dominant root-cause classifications are therefore provisional but informative. For MetroPT and MetroPT2, the current evidence points to alarm-conversion failure and insufficient independent failures: recall can be obtained, but only by issuing many alarms around sparse labelled events. For SCANIA, the dominant issue is estimand mismatch: a vehicle-level repair-risk task is not the same as compressor event warning. For Hydraulic Systems and SECOM, the current high-quantile score finds no useful positive-event detection, consistent with non-event labels and signatures that need different feature families or supervised baselines.

#### 7.4 Score stratification and calibration limits

Figure 3 and Figure 4 are retained as score-stratification diagnostics, not probability calibration evidence. The current model outputs are ranks or anomaly scores. The generated bin tables report mean rank score and observed future-event proportion; the independent positive-event evidence is too limited for a strong calibrated probability claim.

#### 7.5 Metric provenance, baselines, and audit timeline

Table 8 reports real held-out event-level baselines and target-region variants under the same alarm policy as the registered diagnostic. Table 9 isolates the registered score, target-region variants, and negative control. These tables are descriptive rather than a superiority claim: they use only the available held-out failure in each compressor dataset, so precision and false-alarm burden remain the dominant evidence.

Table 10 records the source artifact, split, metric class, value, and independent unit for the real event-level alarm metrics used in this section. The MetroPT and MetroPT2 values are held-out event-level alarm metrics, not timestamp-level classification scores. This prove-

Table 8: Real held-out event-level comparison under the shared alarm policy.

| Data     | Method              | Family           | Target | Alarms | Recall | Precision | FA/day | Lead          |
|----------|---------------------|------------------|--------|--------|--------|-----------|--------|---------------|
| MetroPT  | Flowmeter low       | baseline         | 1      | 9      | 0      | 0         | 0.361  | Not estimable |
| MetroPT  | Max robust z        | baseline         | 1      | 2,527  | 1      | 0.000396  | 101    | 3,346         |
| MetroPT  | Current high        | baseline         | 1      | 2,930  | 1      | 0.000341  | 117    | 3,358         |
| MetroPT  | Oil temp high       | baseline         | 1      | 359    | 0      | 0         | 14.4   | Not estimable |
| MetroPT  | H1 high             | baseline         | 1      | 505    | 0      | 0         | 20.2   | Not estimable |
| MetroPT  | TP2 high            | baseline         | 1      | 2,527  | 1      | 0.000396  | 101    | 3,346         |
| MetroPT  | TP3 high            | baseline         | 1      | 2,503  | 1      | 0.0004    | 100    | 3,330         |
| MetroPT  | Negative control    | negative control | 1      | 300    | 0      | 0         | 12     | Not estimable |
| MetroPT  | Dynamical EVT score | registered       | 1      | 2,527  | 1      | 0.000396  | 101    | 3,346         |
| MetroPT  | Failure prototype   | target region    | 1      | 0      | 0      | 0         | 0      | Not estimable |
| MetroPT  | Rare-state region   | target region    | 1      | 2,527  | 1      | 0.000396  | 101    | 3,346         |
| MetroPT2 | Flowmeter low       | baseline         | 1      | 0      | 0      | 0         | 0      | Not estimable |
| MetroPT2 | Max robust z        | baseline         | 1      | 849    | 1      | 0.00118   | 51.3   | 3,469         |
| MetroPT2 | Current high        | baseline         | 1      | 912    | 1      | 0.0011    | 55.1   | 3,440         |
| MetroPT2 | Oil temp high       | baseline         | 1      | 344    | 1      | 0.00291   | 20.6   | 3,451         |
| MetroPT2 | H1 high             | baseline         | 1      | 664    | 1      | 0.00151   | 40.2   | 2,439         |
| MetroPT2 | TP2 high            | baseline         | 1      | 849    | 1      | 0.00118   | 51.3   | 3,469         |
| MetroPT2 | TP3 high            | baseline         | 1      | 846    | 1      | 0.00118   | 51.1   | 3,442         |
| MetroPT2 | Negative control    | negative control | 1      | 776    | 0      | 0         | 47.1   | Not estimable |
| MetroPT2 | Dynamical EVT score | registered       | 1      | 849    | 1      | 0.00118   | 51.3   | 3,469         |
| MetroPT2 | Failure prototype   | target region    | 1      | 157    | 1      | 0.00637   | 9.41   | 3,435         |
| MetroPT2 | Rare-state region   | target region    | 1      | 851    | 1      | 0.00118   | 51.4   | 3,469         |

Table 9: Target-region variants and negative controls on real held-out event splits.

| Data     | Variant             | Family           | Recall | Precision | FA/day | Metric           | Leakage control                      |
|----------|---------------------|------------------|--------|-----------|--------|------------------|--------------------------------------|
| MetroPT  | Dynamical EVT score | registered       | 1      | 0.000396  | 101    | robust max-z     | train thresholds and references only |
| MetroPT  | Failure prototype   | target region    | 0      | 0         | 0      | robust euclidean | train thresholds and references only |
| MetroPT  | Rare-state region   | target region    | 1      | 0.000396  | 101    | robust euclidean | train thresholds and references only |
| MetroPT  | Negative control    | negative control | 0      | 0         | 12     | robust euclidean | train thresholds and references only |
| MetroPT2 | Dynamical EVT score | registered       | 1      | 0.00118   | 51.3   | robust max-z     | train thresholds and references only |
| MetroPT2 | Failure prototype   | target region    | 1      | 0.00637   | 9.41   | robust euclidean | train thresholds and references only |
| MetroPT2 | Rare-state region   | target region    | 1      | 0.00118   | 51.4   | robust euclidean | train thresholds and references only |
| MetroPT2 | Negative control    | negative control | 0      | 0         | 47.1   | robust euclidean | train thresholds and references only |

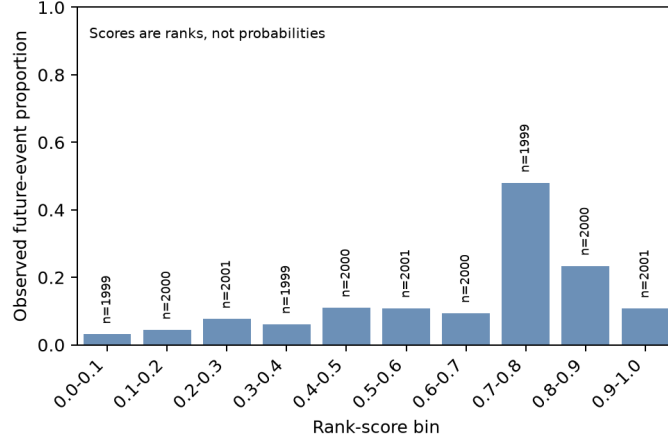


Figure 3: Score-stratification diagnostic for ranked horizon scores. Probability interpretation is not claimed.

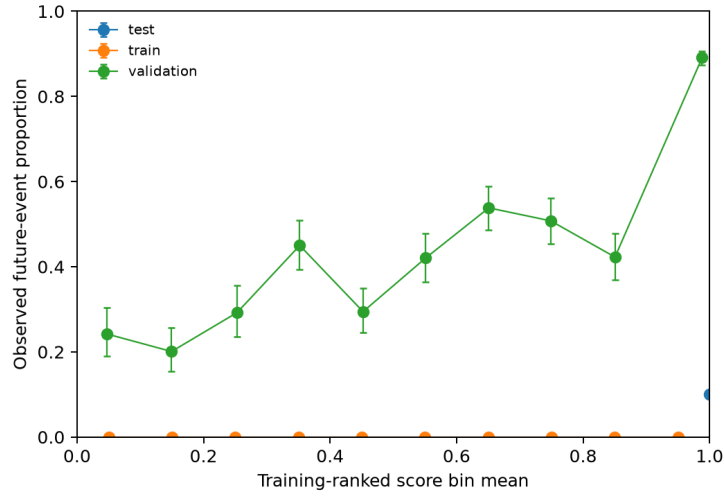


Figure 4: Split-aware score-stratification intervals for ranked horizon scores.

nance prevents event precision, false-alarm burden, and pointwise validation checks from being compared as if they were the same result.

Table 10: Metric provenance for real held-out event-level alarm metrics.

| Dataset  | Evidence type     | Metric               | Value    | Unit            | Use                     |
|----------|-------------------|----------------------|----------|-----------------|-------------------------|
| MetroPT  | event-level alarm | event precision      | 0.000396 | failure episode | real event-level metric |
| MetroPT  | event-level alarm | false alarms per day | 101      | operating day   | real event-level metric |
| MetroPT2 | event-level alarm | event precision      | 0.00118  | failure episode | real event-level metric |
| MetroPT2 | event-level alarm | false alarms per day | 51.3     | operating day   | real event-level metric |

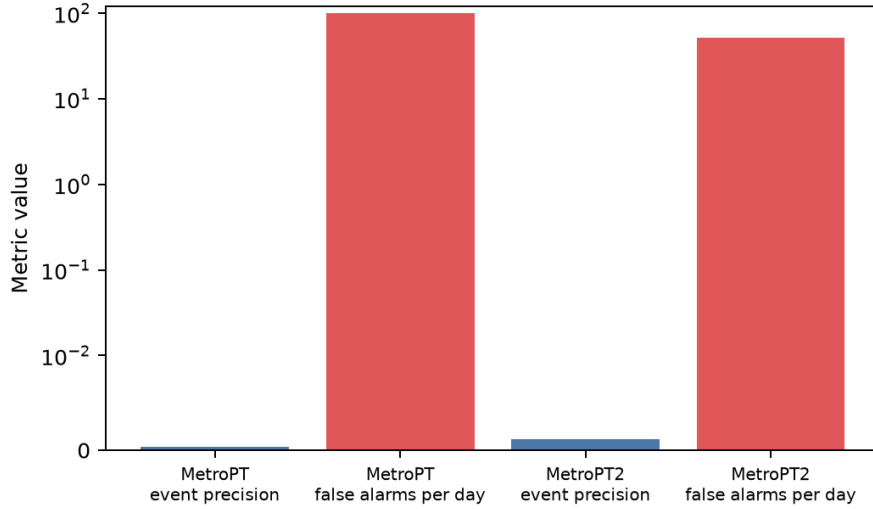


Figure 5: Real event-level metric diagnostic showing near-zero event precision and large false-alarm burden after alarm conversion.

Table 11: Completion status for broader review-requested analyses.

| Requested analysis                           | Status                       | Artifact evidence               | Effect on claims                                |
|----------------------------------------------|------------------------------|---------------------------------|-------------------------------------------------|
| full Monte Carlo validation                  | completed                    | simulation_study_broad.pas      | supports bounded estimator-stability claims     |
| registered dynamical-EVT diagnostic          | completed                    | MetroPT and MetroPT2 event rows | supports negative alarm-burden finding          |
| target-region variants and negative controls | completed                    | event_variant_comparison.csv    | variant claims tied to observed event metrics   |
| real event-level baseline comparison         | completed                    | event_baseline_comparison.csv   | baseline comparison uses the same alarm policy  |
| metric contradiction audit                   | completed                    | metric_provenance table         | pointwise and event metrics separated           |
| real failure timeline                        | completed                    | event_timeline_trace.csv        | timeline is tied to a real held-out event       |
| software archive DOI                         | repository citation complete | CITATION.cff and CodeMeta       | repository URL is cited; DOI is venue-dependent |

Figure 5 visualizes the operational scale of the real event-level metrics: event precision is near zero while false-alarm burden remains large after alarm conversion.

Synthetic baseline-family outputs remain in the supplement as runner audits. The empirical comparison in Table 8 is the event-level evidence. The event timeline in Figure 6 is now generated from the MetroPT held-out test failure; Table 12 records the figure source.

Table 12: Traceability audit for the aligned event-timeline figure.

| Figure                  | Source             | Status    | Missing | Use                    |
|-------------------------|--------------------|-----------|---------|------------------------|
| real_event_timeline.png | MetroPT test event | traceable | none    | empirical timing audit |

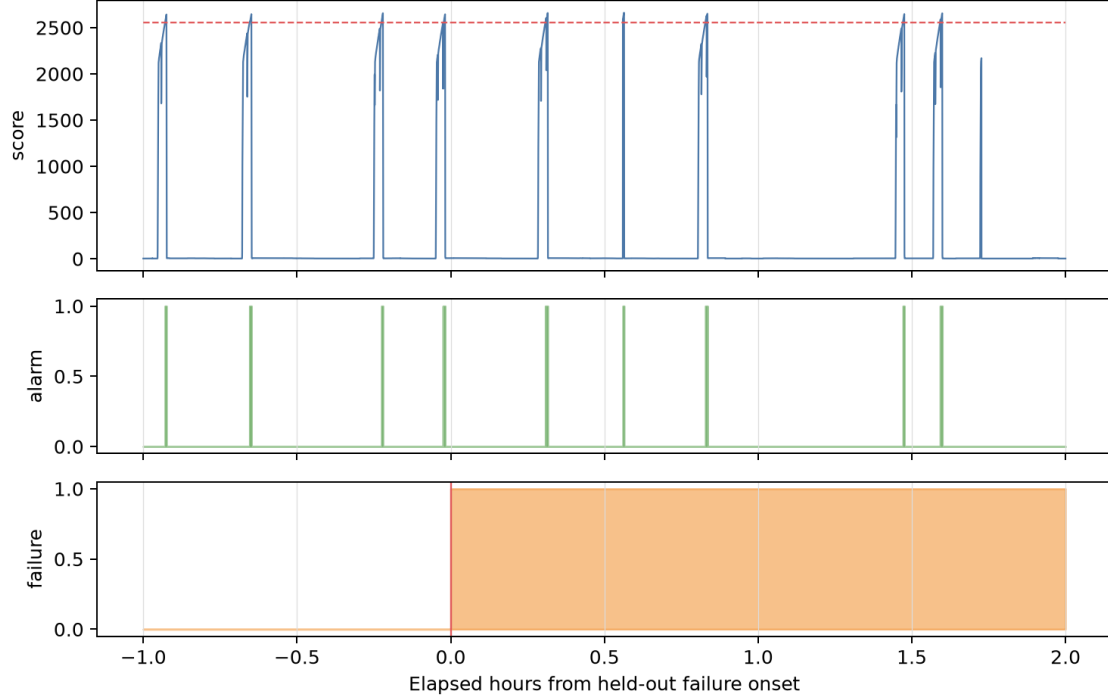


Figure 6: Traceable MetroPT held-out failure timeline. The x-axis is elapsed hours from the registered test failure onset; the panels show the fitted score threshold, alarm flags after thresholding, and labelled failure interval.

## 8 Discussion

The main scientific lesson is negative. Clustered extremes and low extremal-index diagnostics can be observable in industrial telemetry, but that does not imply useful early warning. The conversion from pointwise exceedances to alarm episodes is a separate modelling decision. In the registered threshold grid, that conversion produced a failed-detection region; in the compressor diagnostics, event recall was possible only with thousands of predicted alarm episodes and very low precision.

The likely mechanisms differ by dataset role. For MetroPT and MetroPT2, the limiting factors are sparse independent failures, target-region uncertainty, and alarm burden. A single detected held-out event cannot support a stable estimate of event-level performance when the alarm policy generates many episodes. For SCANIA Component X, the issue is not merely performance; it is a different estimand, because vehicle-level repair-risk ranking is not compressor early warning. Hydraulic Systems and SECOM show another boundary: condition-state and yield labels can be processed by the software, but they do not supply event-onset evidence for dynamical-EVT alarm claims.

The extremal index should therefore be read as a dependence diagnostic. It can identify clustering under a chosen observable, threshold, and run length, but it is not a mechanical degradation variable. Regime mixture, smoothing, missingness, periodic control behavior, and persistent shifts can all create clustered exceedances. Useful deployment would require showing that the target region is identifiable from training data, that failures recur through that region before onset, that enough independent failures or entities exist for uncertainty estimation, and that the alarm policy remains useful after episode conversion.

Calibration is similarly limited. The current outputs are ranks or anomaly scores. Score-stratification plots are useful for detecting orientation mistakes and split drift, but they are not evidence of calibrated failure probabilities unless a separate calibration partition contains

enough independent positive units and the score-to-probability map is tested on held-out data.

The reproducibility layer remains important because it prevents negative results from being silently edited away. The repository checks generated references, citations, unsupported claim language, compiled PDFs, and artifact manifests. These checks do not make the method scientifically successful. They make the boundary between supported evidence and unsupported interpretation explicit.

## 9 Conclusion

The event-level compressor case studies gave a direct negative answer. MetroPT contained one independent labelled test failure; the registered diagnostic detected it, but only with 2527 predicted alarm episodes, event precision 0.00040, and a false-alarm burden of 101.17 alarm episodes per operating day. MetroPT2 also contained one independent labelled test failure; it was detected with 849 predicted alarm episodes, event precision 0.00118, and 51.29 false-alarm episodes per operating day. These detections are therefore not operationally persuasive: the event was recovered, but the alarm burden was far beyond what would constitute a useful maintenance signal.

The statistical interpretation is that clustered extremes are dependence diagnostics, not maintenance decisions. Threshold and run-length choices changed the estimated clustering behavior, which means the extremal-index estimate is conditional on the observable, regime definition, threshold, and declustering rule. Low estimated extremal index can indicate recurrence, persistence, smoothing, missingness, periodic control behavior, or regime mixture; it is not direct evidence of degradation unless the target region is known to be failure-proximal and the episode-level alarm policy remains useful after conversion.

The four research questions are answered narrowly. Regime conditioning and threshold grids exposed sensitivity, but they did not establish a stable diagnostic suitable for general deployment. Declustering and alarm merging did not reduce duplicate alarms while preserving event recall in the registered failed-detection grid; all tested merge-gap and post-onset tolerance combinations had zero recall, precision, and  $F_1$ . The robust extreme-score diagnostic added limited diagnostic value by revealing alarm-conversion failure and estimand mismatch, but it did not add operational early-warning value. The dominant failure mechanisms are sparse independent failures and alarm burden for MetroPT and MetroPT2, vehicle-level estimand mismatch for SCANIA Component X, cycle-state mismatch for Hydraulic Systems, and yield-target mismatch for SECOM.

Recurrence-based EVT can support maintenance decisions only after stronger conditions are met: an identifiable failure-proximal target region, enough independent failures or entities for uncertainty estimation, leakage-free validation, stable regimes and thresholds, fair comparison against simple baselines, acceptable event recall, acceptable false-alarm burden, useful warning lead time, and prospective or external validation. The present study should therefore be read as evidence about the limits of converting clustered extremes into maintenance alarms, not as evidence of predictive superiority.

## Code and Data Availability

Code, configuration files, data-acquisition scripts, generated-paper tooling, and reproducibility materials are maintained in the project repository [9]. Raw third-party datasets are not redistributed by this repository; the acquisition pipeline stores them locally under `data/raw/` and documents their public sources through the dataset registry.

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